Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Healthcare AI and compliance document intelligence engineer with 8+ years building HIPAA-compliant, regulated-data pipelines at production scale. Built **AI on FHIR** — a HIPAA-compliant NLP platform translating natural language queries to FHIR medical record APIs (BERT, LLMs, RBAC, 60% reduction in clinical retrieval time). Built **LeaseI** — an 8-agent LangGraph pipeline for source-cited compliance document intelligence: 100% citation correctness, human-review gate on regulated decisions, 93.6% precision across 1,240 enterprise documents. Built **AgentProbe** — CI-integrated 5-metric LLM-as-judge eval engine, live at github.com/venkatalakshmiparimala/AgentProbe. Shipped Entrata's first production AI pipeline on 1.6M financial records with PHI encryption and zero compliance violations. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

HIPAA-Compliant AI & Healthcare Data Engineering | Source-Cited Compliance Document Intelligence (LeaseI, AI on FHIR) | Multi-Agent Agentic Architecture & HITL Design (LangGraph) | LLM Evaluation & CI Regression Gating (AgentProbe) | Python · SQL · FastAPI · TypeScript | AWS (S3, EC2, RDS, Glue) · Docker · GitHub Actions | Regulated Data Pipelines & PHI Encryption | RAG Pipelines & Vector Search (ChromaDB)

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern

- **Designed and deployed an agentic compliance system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) — every compliance decision grounded in source regulation via RAG (ChromaDB, FAA Part 107 corpus), with cited reasons and validation guardrails blocking non-compliant outputs before export.
- **Engineered validation guardrails and a human-review escalation path** — low-confidence or high-risk decisions route to human review; compliant decisions export automatically with full provenance.

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Achieved 100% PHI compliance** on AWS S3 migration — encrypted 100% of resident sensitive data (phone numbers, SSNs) using Libsodium; zero security incidents. Led Docker-based microservices redesign: 50% scalability improvement, 30% deployment time reduction.
- **Shipped Entrata's first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) — LLM-powered automation with output validation, quality gates, and compliance controls; zero incidents post-deployment.
- **Built compliance-gated data pipelines** for HAP, HUD payments, income eligibility, AMI pricing across 100K+ unit portfolios — zero federal/state violations; optimized 200+ SQL queries (30% performance gain).
- **Delivered 20+ production modules** across 120+ Agile sprints for 100K+ monthly active users — cross-functional delivery with compliance, operations, and product teams; 99.5% uptime over 6+ consecutive months.
- Eliminated 80% of manual operational workload via Python/Bash automation; conducted 300+ peer code reviews; mentored 5+ engineers.

George Mason University

Aug 2024 – Aug 2025

Research Associate / Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed faculty dashboards improving comprehension 40%. Mentored 600+ students in Python and database design.

PROJECTS

AI on FHIR: HIPAA-Compliant NLP for Medical Records

Built a HIPAA-compliant NLP platform reducing manual clinical retrieval time ~60%: translates natural language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforces role-based access control across structured medical records. Deployed on cloud infrastructure with Docker containerization — the same citation-required, regulated-data precision that insurance claim appeals demand.

Flask · React · LLMs · FHIR · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Leasel: Source-Cited Compliance Document Intelligence & 8-Agent HITL Pipeline

Three-surface compliance AI platform sharing one knowledge repository: (1) Document Intelligence — OCR ingestion, field extraction, 100% citation correctness (every answer cited to exact source clause or flagged unsupported), document diff, missing-clause detection, human-review gate on regulated decisions; (2) Support Investigation — 8-agent LangGraph pipeline producing ranked, evidence-backed root-cause hypotheses across business rules, settings, and historical incidents; HITL escalation when confidence too low; (3) Self-Service Routing — document/policy questions answered directly; eligibility/configuration questions route to staff, never resolved or explained by AI. Core rule enforced in code: LLM extracts and reasons; human or deterministic logic makes the final call — the same source-cited, human-gated standard Insurf's appeals product requires. 93.6% precision / 87.9% F1 across 10 compliance rules on 1,240 enterprise documents.

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Multi-Agent Evaluation & LLM Observability Platform Live

3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) and inter-agent state handoffs. CI-integrated 5-metric LLM-as-judge evaluation engine (factual accuracy, hallucination, retrieval, completeness, compliance) — every commit auto-scored, regressions blocked before production. ChromaDB RAG pipeline with trace-level observability.

Python · LangGraph · LangChain · FastAPI · ChromaDB · React · Docker · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, R, Bash

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, HITL Design, LLM Evaluation, LLM-as-Judge, Prompt Engineering, RAG Pipelines, Vector Search, ChromaDB, Embeddings, BERT, spaCy, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, FHIR APIs

ML & Data: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Feature Engineering, NumPy, Pandas, Hugging Face, Tableau

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, Docker, GitHub Actions CI/CD, PostgreSQL, MySQL, ETL, Data Pipelines, Microservices, RabbitMQ, Libsodium (PHI encryption)